

- Q.30 Explain Kirchhoff's voltage law (KVL).
 Q.31 Explain the concept of phase difference in alternating quantities.
 Q.32 Explain energy stored in an inductor.
 Q.33 Explain Thevenin Theorem.
 Q.34 Discuss in detail solar cells.
 Q.35 Difference between electrical and magnetic circuits.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 State and explain Faraday's law of electromagnetic Induction.
 Q.37 What is battery? Explain the construction, principle and working of lead acid battery.
 Q.38 Define power factor. What are the disadvantages of low power factor?

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**2nd Sem / Comp, ECE, IT, I & Control,
 Med. Eltx., Eltx. & Instr., Power Eltx., EEE
 Sub : Basic Electrical engineering**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The resistance of wire varies inversely as _____. (CO1)
 a) Length b) Resistivity
 c) Temperature d) Area of cross-section
- Q.2 The number of cycles completed in one second is called _____. (CO6)
 a) Frequency b) Time period
 c) Voltage d) None of above
- Q.3 The Unit of inductance is _____. (CO7)
 a) Ohm b) Volt
 c) Henry d) Watt
- Q.4 Internal resistance of ideal current source is _____. (CO3)
 a) Very low b) Very high
 c) Infinite d) Zero

- Q.5 The frequency of DC is _____.
 a) 100 b) Infinite
 c) 50 d) zero
- Q.6 Unit of magneto motive force is _____. (CO4)
 a) Amp-turns b) Reluctance
 c) Tesla d) Weber
- Q.7 Power factor for pure inductive load is _____. (CO3)
 a) Unity b) Zero
 c) Lagging d) None of these
- Q.8 The solar cell converts solar energy into _____ energy. (CO5)
 a) Chemical b) Mechanical
 c) Electrical d) None of above
- Q.9 the maximum value of power factor is:
 a) zero b) 2
 c) 1 d) None
- Q.10 Electrical conductivity is measured in:
 a) Mho/m b) Mho/m²
 c) Mho/m³ d) Mho-m

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Draw the symbol of constant voltage source.
 Q.12 Expand MMF.

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- Q.13 Define Time Period.
 Q.14 What are active components?
 Q.15 Define ideal current source.
 Q.16 The wave shape of an AC is _____.
 Q.17 The energy stored in a magnetic field is _____.
 Q.18 KCL stands for _____.
 Q.19 Define Resistance.
 Q.20 What is mutual induction?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Give significance of the power factor.
 Q.22 Difference between A.C. and D.C.,
 Q.23 What is Resonance? Explain parallel resonance in detail.
 Q.24 Explain maximum power transfer theorem.
 Q.25 What is cell? Explain its types.
 Q.26 Discuss average and RMS value of AC.
 Q.27 Discuss active power and reactive power.
 Q.28 Define flue and reluctance.
 Q.29 What are the care and maintenance of Nickel Cadmium batteries?

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